

Nilesh Mishra

+91 8652479323 · mishranilesh968@gmail.com · LinkedIn · Portfolio · GitHub

EDUCATION

S.M. Shetty College (Autonomous), Mumbai
M.Sc. in Information Technology

2023 – 2025
CGPA: 7.2 / 10

S.M. Shetty College (Autonomous), Mumbai
B.Sc. in Information Technology

2020 – 2023
CGPI: 8.18 / 10

PROFESSIONAL EXPERIENCE

Junior Software Developer

May 2025 – Sept 2025

Enirmaan Techsolutions Pvt. Ltd., Maharashtra

- Integrated external APIs for real-time health vitals detection from video; developed a FastAPI backend integrated with a Flutter app, improving patient monitoring efficiency.
- Implemented face validation and multi-face rejection logic to enhance system accuracy.
- Built an OCR-based visiting card extractor using Tesseract, EasyOCR, OpenCV, regex, and Hugging Face LLM with Pydantic validation.

Tech Support Intern

Oct 2024 – Feb 2025

Livingthings.ai (RBTIC, IIT Bombay)

- Analyzed IoT-based AC monitoring data to optimize energy consumption and support sustainability goals.

Desktop Support Intern

Mar 2024 – Jun 2024

Policy Junction Marketing Pvt. Ltd.

- Provided system setup, software configuration, and troubleshooting, reducing internal downtime.

PROJECTS

Health Vitals Monitoring via Video

Jul 2025

- Engineered real-time facial detection and vitals estimation using OpenCV and FastAPI, integrated into a cross-platform mobile app.

OCR-Based Visiting Card Extractor

Jun 2025

- Developed a hybrid OCR pipeline using Tesseract, regex, and Hugging Face LLM with Pydantic schema validation.

Medical Chatbot with RAG

Apr 2025

- Built a context-aware medical Q&A system using LangChain, Pinecone, and OpenAI GPT.

Breast Cancer Detection

Feb 2025

- Designed ML models using tuned KNN and SVM classifiers with high classification precision.

Stock Price Forecasting

Feb 2024

- Implemented LSTM and CNN-LSTM models for time-series stock forecasting with feature engineering.

ONGOING WORK

Non-Intrusive BP Estimation

Jun 2025

- r-PPG signal reconstruction, de-noising, and time-series reassembly using MultiResUNet and UNetDS64.

Contactless Heart Rate Monitoring

Jun 2025

- RGB-based heart rate estimation using red-channel feature extraction and gated refinement mechanisms.

Insurance Report Parser

Jul 2025

- Developing an end-to-end pipeline to extract structured content from insurance PDFs using NLP and layout-aware models.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, HTML, CSS

Frameworks: Flask, FastAPI, Streamlit, LangChain, TensorFlow, PyTorch, Keras, Scikit-learn

Tools: OpenCV, Hugging Face, Pinecone, Tesseract, Pydantic, Git, Docker, AWS

Platforms: Windows, Linux

CERTIFICATIONS & INTERESTS

Certifications: Python for Machine Learning; Exploratory Data Analysis; Introduction to ML on AWS — Great Learning

Areas of Interest: AI in Healthcare, OCR Systems, Generative AI (LLMs, RAG), Computer Vision, Time-Series Forecasting, MLOps

Extracurriculars: Football and Cricket; Singing and Reading